

# AUTOMOTIVE BATTERY MANAGEMENT SYSTEM (BMS) & OT ENVIRONMENT RESEARCH

Name: Rishabh Bhaskar

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## 1. Understanding BMS in Automotive

An automotive Battery Management System (BMS) is an electronic control unit that continuously monitors, manages, and protects rechargeable battery packs in electric and hybrid vehicles. It acts as the vehicle battery's brain, tracking voltage, current, and temperature to ensure safe operation, maximize driving range, and extend lifespan.

### 1.1 Key Functions of an Automotive BMS

- **Cell Monitoring:** Measures individual cell voltage, total pack current, and temperatures across the entire battery assembly.
- **Cell Balancing:** Cells in a battery pack do not always remain at exactly the same electrical state. Cell balancing reduces differences between cells so that individual cells remain within appropriate operating limits and the battery pack can be used effectively. BMS architectures can use passive or active balancing approaches depending on the design.

Passive Balancing: The BMS finds cells with too much voltage and burns off their excess energy as heat through a resistor. It is cheap and simple, but wastes energy.

Active Balancing: The BMS uses tiny circuits (capacitors or inductors) to physically shift energy from a high-voltage cell into a lower-voltage cell. It is highly efficient but complex and expensive.

- **Safety Protection:** The BMS continuously checks for abnormal conditions. Depending on the battery architecture, these can include: overvoltage, undervoltage, overcurrent, excessive temperature, isolation faults, high-voltage system faults

If a serious fault is detected, the BMS can initiate protective actions such as limiting battery operation or opening high-voltage contactors to disconnect the battery.

- **Thermal monitoring & management:** Battery temperature has a major effect on battery performance, safety and lifetime. The BMS monitors battery temperatures and uses this information to maintain appropriate operating limits and coordinate with the vehicle's thermal-management system.
- **Vehicle Communication:** Links directly with the car's primary control computers via internal networks like a CAN bus.
- **State Calculations:** Computes the State of Charge (SOC) for driving range estimates and State of Health (SOH) for long-term battery condition. State of Power (SOP) estimates the amount of power the battery can safely deliver or accept under the current conditions.

### The Core Technology: Estimation Algorithms

Because you cannot plug a sensor inside a battery cell to check its exact chemical energy, the BMS relies on advanced mathematical models to estimate two critical metrics:

#### State of Charge (SOC)

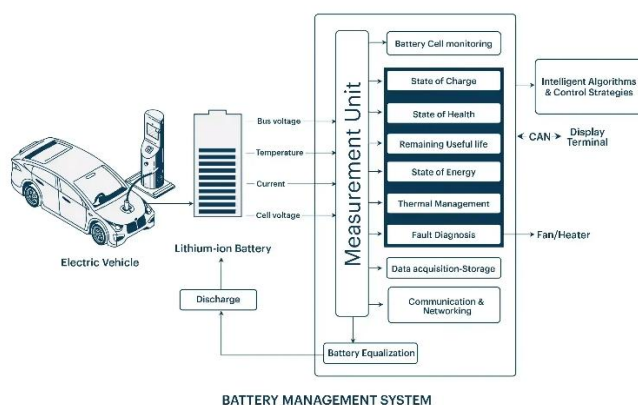
What it is: The battery's "fuel gauge" (0% to 100%).

How it works: It uses Coulomb Counting (tracking exactly how much current flows in and out) combined with Extended Kalman Filters (EKF). The filters map open-circuit voltage against the current to fix errors caused by battery aging or temperature shifts.

#### State of Health (SOH)

What it is: The remaining useful life of the battery compared to when it was brand new.

How it works: It tracks internal resistance growth and capacity fade over time. When SOH drops below 70%, the battery pack is generally considered unfit for automotive use. SOH indicates the battery's health/degradation relative to a reference condition.



A modern automotive BMS is a **distributed safety-critical control system** where slave boards continuously measure individual cells, a master BMS computes SOC/SOH/SOE and safety decisions, communication occurs through protocols such as CAN, CAN-FD, isoSPI, SPI, LIN, and Ethernet, and

the entire system protects, balances, cools, charges, and safely delivers energy from the battery pack to the vehicle.

## 1.2 System Architecture:

An automotive battery pack consists of hundreds or thousands of individual cells. To manage them, the BMS typically uses a **Master-Slave (Distributed) Architecture**:

**Slave Modules (CMUs - Cell Monitoring Units):** These physical circuit boards sit directly on the battery modules. They physically wire to individual cells to measure voltage and temperature, then send this data up the chain.

**Master Controller (BMU - Battery Management Unit):** The main "brain." It collects data from all slaves, runs high-level algorithms (like calculating driving range), controls safety switches (contactors), and communicates with the rest of the car..

## 1.4. Plants where Automotive Battery management systems are manufactured

Automotive Battery Management Systems (BMS) are typically manufactured within specialized EV battery pack assembly plants and electronics component factories run by tier-1 automotive suppliers and cell makers. Major production hubs span global industrial centers in China, South Korea, Europe, the United States, and India.

### Major Automotive BMS Manufacturers & Plant Locations

- **Ottilion Power Systems:** Operates major EV battery and integrated BMS manufacturing plants, including its third Indian facility located in Halol, Gujarat, alongside operations in the US and other global sites.
- **BYD (FinDreams Battery):** Manufactures in-house vehicle batteries and integrated BMS systems across mega-factories in China and expanding production lines in Europe (such as Hungary).
- **LG Energy Solution:** Produces integrated battery packs and advanced BMS architecture via facilities in South Korea, Poland, the United States (e.g., Michigan and upcoming joint-venture sites), and China.
- **Tata AutoComp Systems:** Manufactures EV supply chain components, including battery packs and management systems, through dedicated facilities tied to major Indian automotive hubs.
- **Continental AG & Robert Bosch GmbH:** Produce automotive-grade electronic control units and BMS hardware across multiple component plants situated throughout Germany, Eastern Europe, and Mexico.
- **Battrixx (Kabra Extrusionstechnik):** Develops and manufactures advanced lithium-ion battery packs featuring smart BMS solutions for electric vehicles out of its specialized manufacturing units in India.

## 2. What are the Major Assets & Components?

Operational Technology (OT) refers to the hardware and software used to monitor and control physical industrial processes.

For an automotive BMS manufacturing environment, the major OT assets can be grouped into several layers.

### **Major OT Assets in an Automotive Battery/BMS Manufacturing Plant:**

- **PLCs / PACs**

PLCs (Programmable Logic Controller) are industrial controllers used to control machinery and automated manufacturing processes.

They can control: conveyors, actuators, assembly equipment, robotic cells, safety-related machine functions and production sequences.

PACs (Programmable Automation Controller) provide more advanced control and computing capabilities and may be used in larger or more complex automation systems.

- **Industrial Robots:** Industrial robots can be used for highly repeatable manufacturing operations such as: material handling, assembly, welding, dispensing, inspection, battery-module/pack handling. The exact operations depend on the manufacturing line.

- **Sensors and Actuators**

Sensors collect physical information from the production environment. Eg: temperature sensors, pressure sensors, position sensors, proximity sensors, optical sensors, current/voltage measurement equipment

Actuators perform physical actions based on control commands. Eg: motors, valves, pneumatic systems, robotic actuators

- **HMIs (Human-Machine Interface) :** An HMI provides operators with a visual interface for interacting with industrial equipment.

An operator may use an HMI to view machine status, view alarms, start/stop permitted processes, inspect production information, perform authorised manual operations.

- **SCADA servers (Supervisory Control and Data Acquisition)**

SCADA systems provide supervisory monitoring and control of industrial processes.

They can collect information from PLCs and other devices and provide: real-time process monitoring, alarms, trends, status information, historical data.

- **MES (Manufacturing Execution System):** operates at a higher manufacturing-management level.

**It can be used for:** Tracking production flow, traceability IDs, and quality metrics.

For BMS/battery manufacturing, traceability is particularly important because individual battery components and assemblies may need to be associated with production and test records.

- **AGVs (Automated Guided Vehicle) and AMRs (Autonomous Mobile Robot)**

These systems can transport: components, battery modules, finished assemblies, tools/materials between production areas. They are therefore relevant to the physical logistics side of an automated plant.

- **Battery cyclers / battery test systems:** Used to test battery cells, modules and packs under controlled charge/discharge conditions and validate performance and safety.
- **Industrial Wireless / Wi-Fi**

Wireless technologies can be used for selected industrial applications such as: handheld devices, AGVs/AMRs, maintenance equipment. plant monitoring

However, wireless should not automatically be assumed to be the primary control network for every industrial process.

Critical control functions may use dedicated wired industrial networks depending on the plant architecture.

- **DCS:** Manage continuous chemical or thermal processes in specialized cell-prep areas.
- **Industrial Ethernet/Fieldbus:** Connect shop floor hardware using protocols like PROFINET or EtherCAT.

## 3.1 RIMAC BMS Products

A Battery Management System (BMS) acts as the brain and nervous system of an EV's energy storage. It maintains safety, longevity, and predictable power deployment. Rimac's proprietary Gen.5 BMS architecture relies on a distributed master-slave topology:

### Hardware Units:

- **Battery Management Controllers (BMC - The Master):** The central computing unit. It manages high-level logic, processes isolation monitoring, controls safety contactors, and communicates with the main vehicle control unit (VCU) via high-speed CAN or Automotive Ethernet protocols.
- **Cell Module Controllers (CMC - The Slaves):** Compact satellite boards mounted directly to individual cell clusters. To maximize precision and safety, Rimac integrated highly precise Integrated Circuits (ICs) from Analog Devices (ADI). These ICs measure cell voltages and temperatures simultaneously, passing data up to the BMC via ultra-reliable wired or proprietary wireless daisy-chain communication lines.
- **High-Precision Shunt Sensors:** Custom-calibrated current sensing blocks integrated into the battery's distribution unit. They track micro-amperage fluctuations up to extreme multi kiloamp surges during aggressive acceleration or regenerative braking. Software & Core Algorithms
- **Advanced State Estimations (SoX):** RIMAC's algorithms process physical data in real time to calculate State of Charge (SoC), State of Health (SoH), and State of Power (SoP). They maintain accuracy over the lifetime of the battery, preventing premature cell degradation while letting drivers deploy the absolute maximum safe limit of energy.

- **Active & Passive Cell Balancing:** Features higher-current balancing protocols than standard industry units. When a single cell loses uniformity, the BMS safely dissipates or shifts the excess voltage to recover energy that standard systems leave behind.
- **Functional Safety & Compliance:** Built to strictly adhere to ASPICE Level 2 and ASIL D safety standards—the highest automotive hazard classification achievable. It features rigorous embedded cybersecurity frameworks to prevent unauthorized over-the-air exploitation or malicious hacking of the battery controls.

## 3.2 Where are these products utilized in an automotive scenario?

RIMAC utilizes this hardware and software ecosystem across different automotive tiers, adapting it to handle very distinct deployment challenges:

### 1. Hypercar Extreme-Performance Applications

- **Scenario:** Managing extremely high power and current while maintaining battery safety and thermal stability.
- **Real-World Examples:** Rimac Nevera and Bugatti Tourbillon.
- **The BMS Role:** In high-performance EVs such as the Rimac Nevera and Bugatti Tourbillon, the BMS continuously monitors cell voltage, current, and temperature while estimating the battery's available power. During extreme acceleration, regenerative braking, and high-power operation, it ensures that the battery remains within its safe electrical and thermal operating limits. In the Bugatti Tourbillon, the 800 V battery contains more than 1,500 cells and uses an advanced cooling system. The BMS therefore plays an important role in monitoring the large number of cells and protecting the battery while the vehicle's powertrain demands very high levels of power.

### 2. Premium Mass-Volume OEM Vehicles

- **Scenario:** Providing reliable battery monitoring and control while supporting long-term durability, fast charging, and efficient thermal management.
- **Real-World Example:** All-electric BMW i7.
- **The BMS Role:** Rimac Technology and BMW Group jointly developed a high-voltage battery system for the BMW i7. The system combines BMW's Gen6 battery-cell technology with Rimac Technology's battery-system expertise and Gen5 module-based architecture. The BMS continuously monitors the individual battery cells and the overall battery system, including voltage, current, and temperature. This allows the system to manage charging and discharging safely, maintain the battery within its appropriate operating conditions, and support the vehicle's performance and durability requirements.

### 3. Automobili Pininfarina Battista

- **Scenario:** High-performance electric hypercar requiring extremely rapid power delivery, precise battery monitoring, and thermal protection during sustained high-power operation.
- **Real-World Example:** Automobili Pininfarina Battista.

- **The BMS Role:** The Battista uses a high-performance battery and powertrain derived from technology developed within the Rimac ecosystem. The BMS continuously monitors cell voltage, temperature, current, and overall battery condition to determine how much power can safely be delivered or accepted by the battery. During hard acceleration, regenerative braking, and high-performance driving, the BMS works with the vehicle's powertrain control systems to maintain the battery within safe electrical and thermal limits. This allows the Battista to deliver very high instantaneous power while protecting the cells from over-voltage, over-current, excessive temperature, and other conditions that could accelerate degradation or create safety risks.